

It’s the Elicitation, Not the Eyes: Diagnosing VLM Failures in Slide Quality Inspection and Routing a Symbolic–Neural Critic

Ruihan Su*

Sun Yat-sen University

Abstract

Vision–language models (VLMs) are the default critic inside slide- and document-generation agents, yet they routinely pass slides whose text visibly overflows its box. Is this a failure of *perception* (the model cannot see the defect) or of *calibration* (it sees but will not commit)? The answer decides whether the remedy is a larger model, a different prompt, or a different tool. Our primary contribution is an attribution protocol that settles this per defect, built on a *lossless structured-geometry oracle*—element boxes, text, and style, not a model-written caption that re-injects perceptual error.

The diagnosis is defect-dependent. For a constructed render-overflow class G7 (declared box legal, rendered pixels overflow) the failure is pure *format suppression*: the frozen model that abstains under the conventional pointwise+rubric prompt detects the defect once it is merely *named*, and an atomic elicitation recovers it across four of six VLMs and transfers to human-annotated real slides. For declared geometry (G1) and image/text contradiction (S6) the gain instead needs a paired clean reference—a milder *availability-of-reference* effect. Genuine floors remain that no elicitation rescues: a sub-threshold acuity tail, OCR-bound terminology, and G6 global content position.

The diagnosis prescribes the critic, not the reverse. Declared geometry routes to a symbolic linter and the format-suppressed classes to one re-elicited VLM, lifting coverage from **2/9** classes (the conventional pointwise VLM) to **8/9** (balanced accuracy 0.89) and matching a pre-registered routed hybrid—so the hybrid is a *consequence* of the diagnosis, not an architecture we advocate. A reward-side audit of five published scorers shows the same split: a linter, a document reward, and an artistic-aesthetic head miss G7, while both general-multimodal rewards and a zero-shot perceptual-quality probe catch it—so G7 needs a perceptually-capable engine, not a specialised one. The diagnosis reproduces on *real* CC-licensed layouts via a lossless tool-extracted oracle. The full hybrid targets IR-owning generation agents; for bare third-party pixels the deployable critic is the re-elicited VLM.

1 Introduction

An automatic critic that flags layout and content defects is now a load-bearing component of every slide- and document-generation agent, and the default critic is a general-purpose VLM asked to grade a rendered slide. These critics fail in a revealing way: shown a slide whose title visibly spills out of its box, a strong VLM under a standard rubric prompt confidently answers “no defect.” Whether this is a failure of *perception* (it cannot see the overflow) or of *calibration* (it sees but will not commit) is rarely asked—yet the answer decides whether the remedy is a larger model, a different prompt, or a different tool entirely. We treat that question as the design problem and answer it empirically.

The central observation. A VLM that “cannot see” a slide defect is hiding two failures of opposite character, and separating them is the whole game. Some defects are *genuinely sub-perceptual*: a three-pixel alignment offset or a one-point font mismatch sits below the model’s effective acuity, and no amount of prompting, scale, or resolution recovers it. Others are *format-suppressed*: the model can perceive them, but the conventional elicitation—“here is a slide and a rubric; score every defect type in one pointwise pass”—buries the signal, and merely *changing how we ask* (a forced choice between two candidates, or an atomic yes/no per defect type with a required piece of evidence) recovers it with the model frozen. Conflating the two is why the field oscillates between “VLMs are bad at layout” and “VLMs can judge slides”—both hold, for different defects.

*Independent technical report. All experiments were run on a single 4×RTX 3080 (20 GB) workstation; weights, renders, and raw rollouts are not redistributed but all analysis scripts and reports are released. Correspondence: surh6@mail2.sysu.edu.cn.

Diagnosis -> architecture: route each defect to its bottleneck-appropriate engine

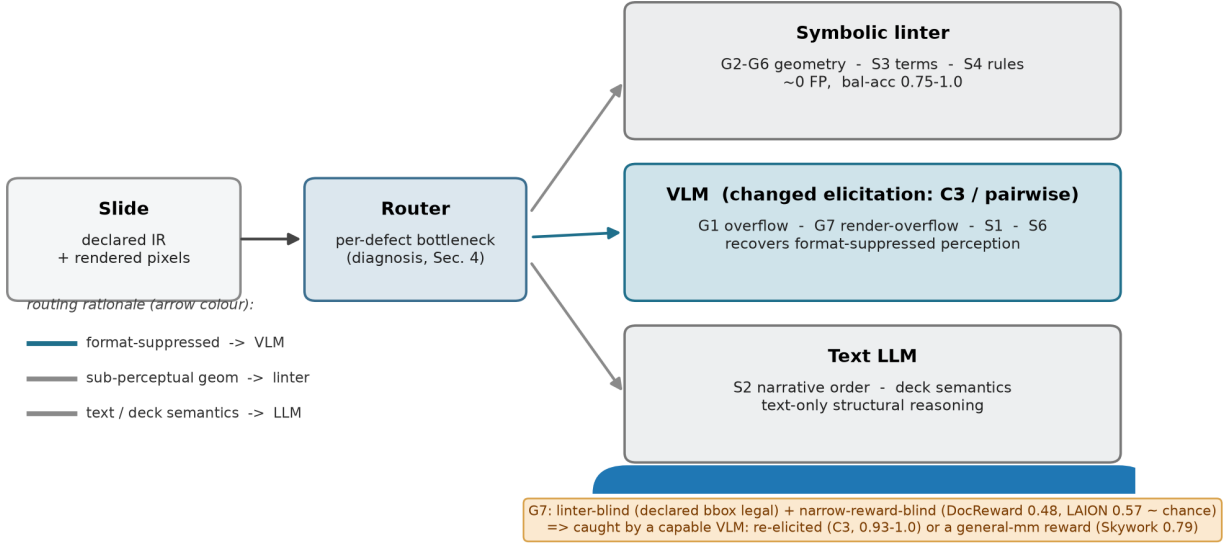


Figure 1: The paper in one picture. A perception/reasoning diagnosis (Sec. 4) shows slide-inspection failures are of two kinds—declared-geometry (exact and cheap for a symbolic linter, plus a sub-threshold acuity tail nothing recovers) and merely format-suppressed (route to a VLM under a changed elicitation). Routing each defect to its bottleneck-appropriate engine yields a hybrid critic (Sec. 5). The render-overflow class G7 is the crux: a symbolic linter is blind to it by construction and a document reward and an artistic-aesthetic head are blind empirically, while *both* general-multimodal rewards (two backbones), a perceptual-quality probe, and the VLM engine under atomic-binary elicitation catch it—so G7-detection tracks perceptual capability, not the domain label, and G7 routes to a perceptually-capable engine.

Diagnosis with a lossless oracle. We make the split measurable with an attribution protocol built on a *structured-geometry oracle*. Each slide carries an intermediate representation (IR) of element bounding boxes, text, and style; we present the same paired (defective, clean) slides under four input modalities—image-only, structured-oracle, VLM-caption, and image+oracle—crossed with detect/localize/repair tasks. Because the oracle is lossless machine-readable structure rather than a model-written caption, the perception channel it isolates is clean: a model that fails image-only but succeeds with the structured oracle is bottlenecked on perception, not reasoning. This is the methodological difference from concurrent perception/reasoning attribution that uses natural-language perception oracles, which re-inject the very perceptual error they try to factor out.

From diagnosis to the critic. The diagnosis—not an architectural preference—prescribes the critic (Fig. 1): route each defect to the engine that owns its bottleneck. Sub-perceptual geometry goes to a symbolic linter that reads coordinates and rules at near-zero false-positive rate; text and structural semantics go to a language model; and the format-suppressed, perceivable classes go to a VLM, but only under the changed elicitation the diagnosis shows recovers them. To stress-test the design we construct an adversarial class, G7 render-containment overflow, that is invisible to the linter by construction (the declared box is legal; only the rendered pixels overflow) and—we find—invisible to a document reward and an artistic-aesthetic head as well, while *both* general-multimodal rewards (distinct backbones) and a zero-shot perceptual-quality probe see it; G7-detection tracks perceptual capability rather than the training-domain label, isolating exactly the slice of the problem that needs a perceptually-capable engine (prompted VLM or reward head).

Contributions.

- **C1 – A lossless-oracle attribution method (the paper’s core contribution).** A perception/reasoning attribution protocol built on a *lossless structured-geometry oracle*—element boxes, text, and style, not a model-written caption that re-injects perceptual error—that separates a VLM’s quality-inspection failures into *genuinely sub-perceptual* versus *format-suppressed* (Sec. 4) and reproduces on *real* CC-licensed slide layouts via a tool-extracted lossless

oracle, with no human or self-annotation (Sec. 7). This measurement—which bottleneck a VLM hits on a given defect—is the durable, domain-general contribution; the critic below merely follows from it.

- **C2 – Format suppression, not capability—scoped to where it holds.** A change of *elicitation alone*, with the model and image frozen, recovers detection that the conventional pointwise+rubric format suppresses. The mechanism holds at *full strength for the constructed render class G7* (and its atomic-binary C3 mirror), where *naming the defect alone*—a single slide, no reference—recovers it; for declared geometry G1 and image/text contradiction S6 the gain instead needs the paired clean reference, a milder *availability-of-reference* effect we label as such. The G7 recovery replicates across the capable models (four of six VLMs, four families, three scales), transfers to human-annotated real slides, and survives a forced-evidence localization check that rules out a lucky “yes” (Sec. 5).
- **C3 – A minimal hybrid critic that follows from the diagnosis, and a falsifiable linter-blind class.** Because the diagnosis says declared geometry is linter-owned and the format-suppressed classes are VLM-recoverable, re-eliciting the *same* frozen VLM (C0→C3) and routing declared geometry to a symbolic linter lifts coverage from **2/9** classes (the conventional pointwise VLM) to **8/9** (balanced accuracy 0.89)—matching a pre-registered routed hybrid (8/9, 0.88) while a control shows per-class prompt/LLM routing is *not* load-bearing. The render class G7 (declared box legal, rendered pixels overflow) makes the linter’s blind spot falsifiable; auditing five published reward scorers over it shows the same perception-capability split—a symbolic linter (0.00), a document reward (0.48), and an artistic-aesthetic head (0.57) miss it, while *both* general-multimodal rewards (0.79, 0.71, distinct backbones) and a perceptual-quality probe (0.83) catch it, so G7 needs a perceptually-capable engine, prompted or reward-headed (Sec. 5).
- **C4 – A specialist examiner and a perturbation-fidelity audit (supporting).** An 8B examiner finetuned on injected defects surpasses a zero-shot 30B on *in-distribution* semantic inspection and learns to abstain rather than hallucinate geometry; separately, 45% of injected geometry defects never render under a standard template renderer—a hazard any IR-labeled slide dataset inherits (Secs. 6, 5).

The full symbolic–neural critic is built for IR-owning generation agents—the setting where the declared structure the linter relies on is available—and for bare third-party pixels, where that structure is missing, the deployable critic is the C3-elicited VLM (Sec. 7). This is a clear deployment contract, stated up front; its boundaries (a small downstream effect, a sim2real gap on bare pixels) are scoped in Sec. 8, not advanced as results.

2 Related work

The VLM perception bottleneck. A growing line of work argues that VLM failures on visually grounded tasks are dominated by perception rather than reasoning—decomposing the bottleneck for chart understanding [1], or showing that the needed visual information is present in the representation but unused [2]; MMVP [3] is the canonical demonstration that VLMs stumble on minimally-different image pairs humans solve trivially—the sub-perceptual acuity floor our title (“not the eyes”) deliberately echoes. We share the perception-first stance but ask it of a different task family—*quality inspection of generated artifacts*—where the perception channel can be isolated exactly with structured geometry rather than approximated by captions, and where the answer feeds a critic architecture rather than a benchmark.

Perception/reasoning attribution. Concurrent work separates perception from reasoning in abstract visual-reasoning benchmarks via natural-language perception oracles [4]. We differ in three ways: (i) our domain is inspection of generated documents, which admits a controllable defect taxonomy with *exact injected labels*; (ii) our perception oracle is a *lossless structured representation*, not a lossy caption, yielding a cleaner attribution boundary; and (iii) we close the loop from diagnosis to a deployed critic. Layout-error-detection work [5] pioneers distribution-grounded structural-error injection but for *document-layout-analysis predictions*, and explicitly notes that existing metrics cannot isolate whether failures stem from recognition or reasoning—precisely the gap our attribution protocol fills.

Relative vs. absolute judgement. Our finding that relative elicitation beats absolute scoring echoes work showing VLM judges can rank but cannot reliably score [6]; that result concerns a judge’s *output mode*, orthogonal to our *source of failure*, and we adopt it to motivate the pairwise and atomic-binary heads of our critic. More broadly, comparative/relative elicitation is known to recover discriminative signal that pointwise scoring suppresses in LLM/VLM judges [7], even though the protocol choice is not bias-free and can itself distort preferences [8]; our contribution is not that effect but its *defect-class boundary*—it rescues G1/S6 yet not G3/S3—which is what turns the elicitation effect into a perception/format attribution signal.

Table 1: Defect taxonomy. The geometry group is detected symbolically; the semantic group is the examiner’s domain; G7 is constructed to be linter-blind. “Channel” is the engine the diagnosis assigns (Sec. 4–5).

Group	Class	Description	Channel
Geometry	G1 text overflow	text exceeds its box	VLM (pairwise) + linter
	G2 element overlap	boxes intersect	linter
	G3 alignment offset	small mis-alignment	linter
	G4 font-size inconsistency	off-grid font size	linter
	G5 brand-color violation	off-palette color (ΔE)	linter
	G6 margin violation	element past safe margin	linter
Semantic	S1 title/body mismatch	body contradicts title	VLM
	S2 narrative-order break	deck sections out of order	LLM
	S3 terminology inconsistency	term drift across deck	term linter
	S4 density violation	slide over/under-packed	LLM
	S5 missing logic section	dropped argument step	LLM
	S6 image/text contradiction	figure contradicts caption	VLM
Render	G7 containment overflow	legal box, rendered overflow	VLM (atomic-binary)

Slide generation, evaluation, and reward. Slide and poster generation with design feedback is an active area; aesthetic-reward systems [9, 10] report that iterative VLM refinement helps gross layout damage but fails on fine-grained aesthetics and is prone to reward hacking, and slide-evaluation studies [11] call for calibrated critic-in-the-loop selection. These motivate, rather than compete with, a routed critic: they observe the VLM’s perceptual ceiling and route around it with rules, whereas we first *diagnose* that ceiling and then show part of it is recoverable by elicitation. We use the human-annotated SLIDEAUDIT set [12] as our real-data probe and audit *five* published reward scorers (two per audited category)—the document reward DocReward [13]; two general-multimodal rewards on distinct backbones, the value-head Skywork-VL-Reward [14] and the CLIP-contrastive PickScore [16]; and two aesthetic/perceptual scorers, a LAION aesthetic CLIP predictor [15] and the zero-shot CLIP-IQA quality probe [17]—as neural-critic baselines.

Critic / reward-model training. Training specialist critics from paired data is established for natural images and code; our examiner adapts the recipe to the rendered-document domain, where defect injection yields exact labels at zero annotation cost. We are careful *not* to claim injection itself as novel—it is a known recipe—only its use as the semantic engine of a routed critic and the perception/reasoning attribution it enables.

3 Problem setup and defect taxonomy

Representation and taxonomy. Every slide is an IR of typed elements with bounding boxes, text, and style; defects are injected programmatically into the IR and then rendered, giving exact labels and matched (defective, clean) pairs. The taxonomy (Table 1) has a *geometry/perception* group G1–G6, a *semantic/reasoning* group S1–S6, and one constructed render class G7; the taxonomy is aligned to the expert-derived SLIDEAUDIT taxonomy [12] via a bidirectional map (results are reported in its canonical classes), with G7 refining its “content overflow/cut-off” category. We add G7 because it is the sharpest test of the linter’s blind spot: its declared box is legal (inside the safe margin, no box overlap) but the rendered content overflows its container, so the defect exists *only* in pixels.

Attribution modalities and metric. We present each paired sample under four modalities: **A** image-only; **B** the lossless structured oracle (with ground-truth markers stripped by a single de-leaking view, regression-tested); **B’** a VLM-written caption held fixed across models; and **C** image+oracle—crossed with detect/localize/repair. The attribution logic is: $A\text{-fail} \wedge B\text{-succeed} \Rightarrow \text{perception bottleneck}$; $A\text{-fail} \wedge B\text{-fail} \Rightarrow \text{reasoning bottleneck}$; $A\text{-succeed} \wedge \text{repair-fail} \Rightarrow \text{execution bottleneck}$. Throughout, the headline metric is *balanced accuracy on paired clean controls*, never recall alone—a consistency defect scored by recall reads a false “100% detection” when the model simply always flags.

Two linters. A geometry linter detects G1–G6 from coordinates and rules (brand color via CIELAB ΔE_{2000} , not RGB distance), at zero false positives on clean slides. A terminology linter handles S3 from an occurrence table plus

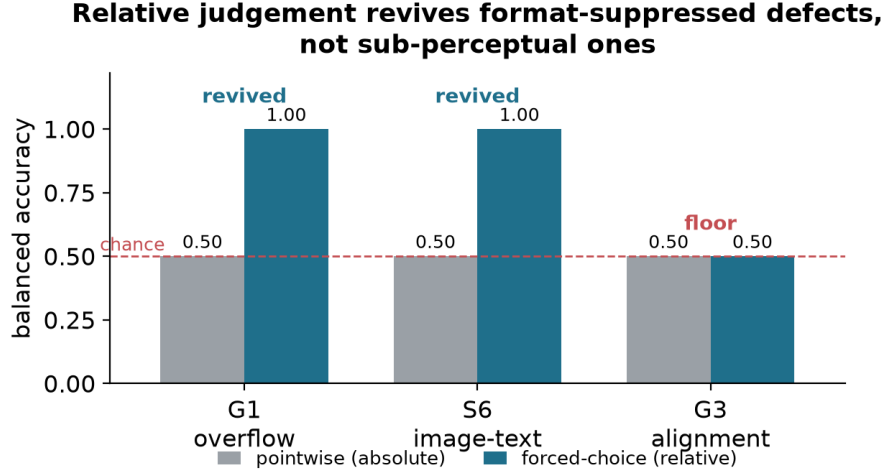


Figure 2: Relative judgement recovers absolute-pointwise blind spots. For G1 overflow and S6 image/text contradiction, pointwise (absolute) inspection is at chance but a forced choice recovers perfect detection (*elicitation-recoverable*). G3 alignment behaves the same way above a magnitude threshold (forced choice 1.00 once the offset clears $\approx 0.8\%$ of slide width; Fig. 3); only its sub-threshold tail and S3 terminology stay at chance side by side. Same model, same images; only the elicitation changes. What *drives* the recovery (naming vs. a paired reference) is decomposed in Fig. 6.

near-duplicate clustering, image-free. Both are the “detectors of record” the diagnosis routes to.

4 Diagnosis: two kinds of blindness

Pointwise image-only inspection floors fine geometry. Under pointwise image-only inspection, G2–G6 sit at chance for 4B and 8B models, and a 30B model breaks only the coarsest class (G1 overflow). This pointwise floor is invariant to the things one would tune first: swapping the vision encoder across *five* families (SigLIP2, an LLM-based encoder, InternViT, a native-resolution NaViT, and MoonViT) leaves every model at the random line, differing only in abstain-vs-over-flag bias, and raising input resolution from 1536 to 2048 changes nothing. The lever is the elicitation and the defect magnitude, not the encoder or the pixel budget (the next paragraph and Fig. 3 show targeted/relative elicitation lifts supra-threshold alignment well above chance)—so the floor is a property of the *pointwise* protocol, not a hard capability limit. Either way the right detector for declared geometry is the symbolic linter: it reads the coordinates directly at balanced accuracy 0.75–1.0 with zero false positives—cheaper and exact where the IR is available, not because the VLM cannot see the defect.

But some “perceptual” failures are elicitation artifacts, not blind spots. The same pointwise image-only setting hides a second, opposite phenomenon. For G1 text overflow and S6 image/text contradiction, pointwise balanced accuracy is 0.50—indistinguishable from the genuinely sub-perceptual classes—yet a two-alternative forced choice between the defective and clean slide jumps to **1.00** (Fig. 2): the defect is perceivable, and the absolute pointwise call simply fails to elicit it. The same is true of G3 alignment, but only *above a magnitude threshold*: when we sweep the injected offset as an internal contrast (one element shifted out of its aligned sibling column, decidable from the slide alone) across 2–64 px (0.1–3.3% of slide width), the forced choice saturates to **1.00 by 16 px** (0.83%) and single-slide atomic elicitation climbs to a ≈ 0.90 plateau by 48–64 px, while the *sub-threshold tail* (≤ 8 px) stays at chance even side by side (Fig. 3). So fine alignment is not a class-level capability floor: it is format-suppressed and recoverable wherever the offset clears acuity, with a genuinely sub-perceptual residue only in the small-offset tail. (*How* the elicitation helps differs by class—a named target on a single slide vs. a paired clean reference—and we decompose it in Sec. 5.1 (Fig. 6), finding pure format suppression for the constructed render class G7 and a milder availability-of-reference effect for G1/S6.) The connective thread is therefore a *magnitude-gated* recoverable-versus-sub-perceptual boundary rather than a clean geometric/semantic split—and S3 terminology (below) remains the one defect no elicitation rescues.

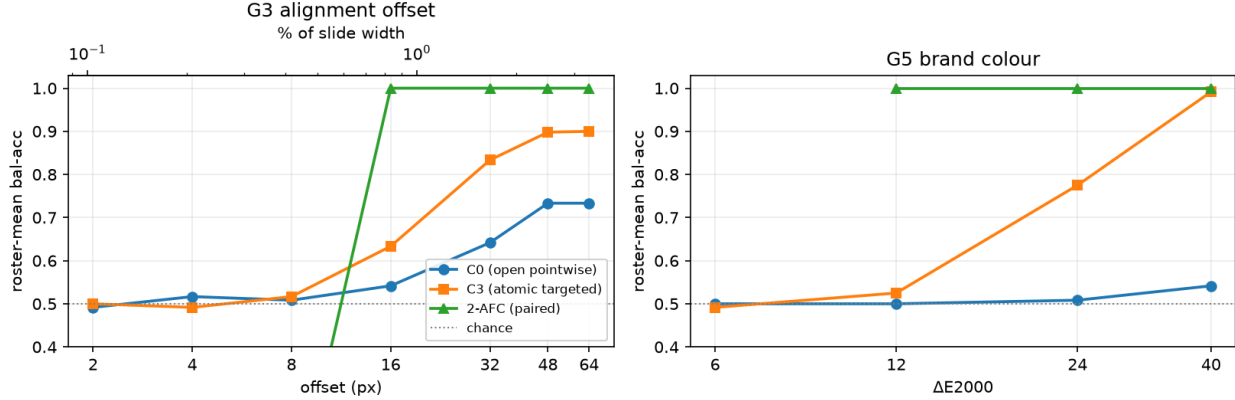


Figure 3: Fine geometry is magnitude-gated, not a flat capability floor. Roster-mean balanced accuracy (6 VLMs, matched-twin internal-contrast set) versus injected magnitude for G3 alignment (left; px and % of slide width) and G5 brand color (right; ΔE_{2000}). Open pointwise (C0) lags; atomic targeted elicitation (C3) recovers to a ≈ 0.90 plateau by 48–64 px / 0.99 by ΔE 40; the paired 2-AFC saturates to 1.00 by 16 px (0.83%) / ΔE 12. The sub-threshold tail (≤ 8 px, $\leq 6 \Delta E$) stays at chance under every protocol—the genuinely sub-perceptual residue. Single-slide C3 plateaus below 1.0, a residual ceiling on *absolute* spatial judgement that contrastive elicitation does not share.

A counterexample: terminology. Not every consistency defect is format-suppressed. S3 terminology inconsistency is *not* rescued by forced choice (0.625, heavy position bias): reading the same term across deck pages and spotting a subtle variant is below threshold even side by side. Tellingly, the structured-oracle channel—which feeds the model the full deck text with both variants present—also fails (0.56), so the bottleneck is OCR-from-pixels plus the yes/no framing, not reasoning. S3 therefore leaves the VLM entirely for a symbolic terminology linter, which reaches balanced accuracy 1.000 against the VLM’s 0.69—a negative result that *re-routes* the class.

A genuine geometric blind spot: G6 page offset. The re-operationalisation does not rescue every class; on one it fails cleanly. We re-pose G6 margin as a *page offset*: the whole content block is shifted toward one edge—every element together, so internal alignment is preserved and the defect cannot be misalignment—leaving one margin crowded against (or running off) the edge and the opposite side empty, decidable from the slide alone.¹ Across all six VLMs and both pointwise framings (absolute “past the safe margin” and relative “shifted to one side”) the page offset stays at chance (roster-mean C3 0.50), and the discriminative 2-AFC—the very test that separated *data artifacts* (S6 image/text contradiction, which recovers to perfect once the figure is actually rendered) from genuine blindness—leaves G6 on the floor for *every* model (Fig. 4): none can tell a page-offset slide from a balanced one until the content is clipped by the page boundary. So unlike G3/G5 (format-suppressed, recoverable) and S6 (a rendering artifact, recoverable), page offset is a *genuine* perceptual blind spot—VLMs do not represent global content positioning—which is exactly why declared geometry routes to the linter.

A methodological aside that becomes load-bearing. Enterprise “snap-to-master” templates re-fit element boxes to a grid before rendering. Measuring this, we find a template silently *absorbs* 45% of injected geometry defects: overlap, alignment, and margin violations become pixel-identical to their clean twins after snapping (Sec. 5). This collapses the inspectable spectrum toward two poles—pure-perception defects that templates cannot constrain (overflow, driven by content-determined text length) and pure-semantic defects that templates do not touch—and it is a trap for any dataset that derives labels from the IR but evaluates on the rendered image. We use it below to explain a reward-model failure and to justify rendering defectives faithfully.

¹The original S6 render omitted the figure element, so the cross-modal contradiction was literally unseeable (recall 0); on the valid-figure corpus the S6 2-AFC reproduces at 0.75–1.00 across all six VLMs, confirming it is a render artifact, not blindness.

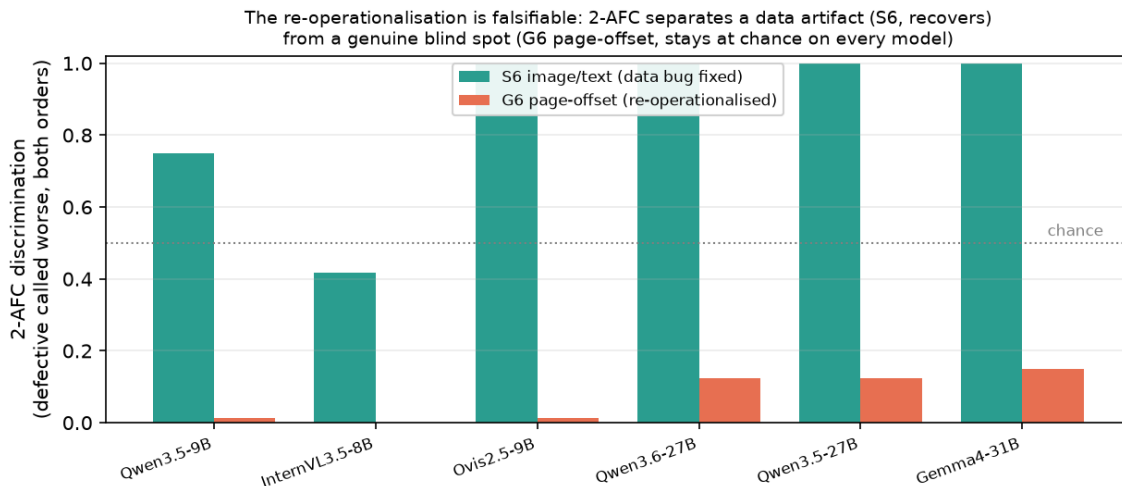


Figure 4: A genuine blind spot the forced choice exposes. Per-model 2-AFC discrimination (fraction of paired slides where the defective is consistently called worse in both presentation orders; ties count as not-discriminated). Single-image pointwise puts both classes at chance, but the forced choice *separates* them: the data artifact S6 (image/text contradiction, on the fixed valid-figure corpus) recovers on every model, while the re-operationalised G6 page offset stays at the floor on every model—a genuine blind spot, not an elicitation or data artifact.

5 From diagnosis to a symbolic–neural critic

The diagnosis says a single model is the wrong unit: route each defect to the engine that owns its bottleneck. We build that critic, validate the elicitation recovery it relies on across many models, and stress it with the constructed render class G7.

5.1 Elicitation: format suppression is not capability

Four ways to ask. We compare four elicitations holding the model and image fixed: **C0**, the conventional pointwise+rubric pass over the whole taxonomy in one call; **C1**, free-form describe-then-classify; **C2**, a geometry-normalized synthetic-twin pairwise (re-render snapped-to-grid, both orders); and **C3**, an atomic per-type yes/no with forced localization. The scientific contrast is **C3 vs. C0**: same model, taxonomy, and image, differing only in “ask everything at once” versus “one atomic binary with forced evidence.” If $C3 \gg C0$, the pointwise abstention was a format/cognitive-load artifact, not a capability gap.

The recovery replicates across families and scales. On the constructed render class G7, the atomic-binary C3 recovers detection that C0 suppresses, and it does so across four model families and three scales (Fig. 5): Qwen3.5-9B $0.50 \rightarrow 0.93$, Qwen3.5-27B $0.93 \rightarrow 1.00$, Qwen3.6-27B $0.52 \rightarrow 1.00$, and Google Gemma4-31B $0.75 \rightarrow 1.00$, all at precision $0.88\text{--}1.00$. As a *cross-model* effect—the claim we actually make—the recovery is overwhelming: a stratified (Cochran) McNemar over the four models’ matched pairs counts **149** G7 items that C3 fixes against **zero** reversals ($\chi^2_1=149$, $p < 10^{-30}$).² That the *same* effect appears in models from different vendors with different pretraining is the core model-agnostic evidence: C0’s fixed taxonomy cannot *name* an off-taxonomy render class, so it abstains; the atomic per-type binary restores it.

It is real perception, not a lucky “yes.” Paired-clean precision near 1.00 already rules out an always-flag model, but to confirm a correct “yes” points to the right place we check C3’s forced evidence against the synthetic G7 ground truth. On the four capable models’ G7 true positives, the named region and the named overflowing *element* are correct $\geq 98\%$;

²Per model the gain is individually significant for the three low-C0 models ($p \leq 2 \times 10^{-9}$); on the already-strong Qwen3.5-27B (C0 0.93) only 9 items can move and all 9 go to C3 (0 reversals, exact $p=0.004$, individually n.s. after Holm over the full 61-test family)—a ceiling effect, not a fragile one. Holm (FWER) over all 61 reported tests keeps the pooled G7 recovery, the three low-C0 G7 cells, both examiner contrasts ($p_{\text{Holm}}=1.9 \times 10^{-5}$, 4.7×10^{-4}) and *both* general-multimodal rewards’ G7 detection (Skywork $p_{\text{Holm}}=2 \times 10^{-6}$, PickScore 2.6×10^{-3}); full corrected table + the stratified test in the released multiplicity report.

Atomic-binary elicitation (C3) recovers the linter-blind render class G7 across 4 model families and 3 scales

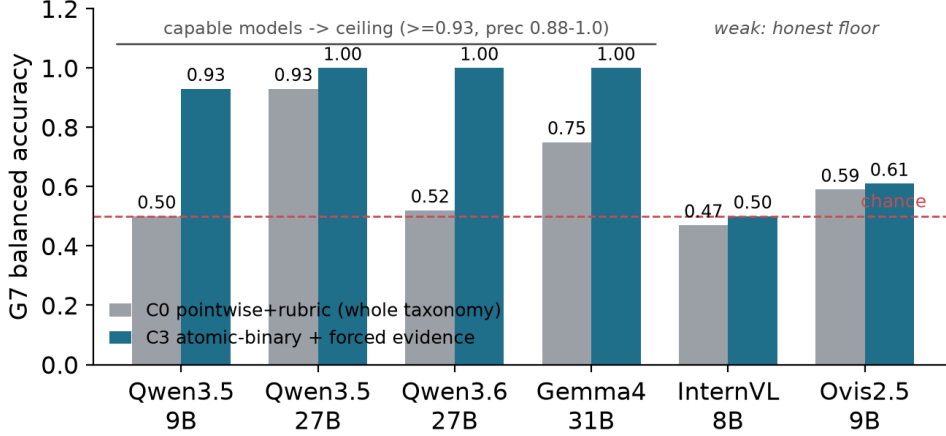


Figure 5: Atomic-binary elicitation (C3) recovers the linter-blind render class G7 across families. Capable models reach the ceiling (≥ 0.93 , precision 0.88–1.0); the weaker InternVL/Ovis stay near their perceptual floor—a capability-dependent boundary, not a universal fix.

models read back the specific spilling content (“the list items starting from ‘Cost attribution’”), which a bluffing model cannot do. The recovery is perception, not a calibration trick on the label.

No free lunch: elicitation is conditional. The optimal elicitation is jointly determined by model capability and defect class. C2 synthetic-twin pairwise—not C3—is what rescues *declared* geometry G1 on capable models (Qwen3.5-27B 0.69 $\rightarrow$ 0.93, Qwen3.6-27B 0.53 $\rightarrow$ 0.78, precision 1.00), because snapping absorbs the declared overflow and the pairwise contrast exposes it. And C1 free-form, which rescues weak abstaining models, *collapses* on strong models: the capable, verbose models free-associate nits and flag a defect on nearly every clean slide (precision ≈ 0.50). There is no universal elicitation—which is precisely why the critic routes per defect and per model rather than adopting one prompt.

Naming vs. a paired reference: a defect-dependent answer. A skeptic asks whether the chance $\rightarrow$ 1.00 recovery is genuine *elicitation* (the model perceives the defect and the format suppressed it) or merely an *easier task* (the forced choice hands over a clean reference and a named target the pointwise call withholds). We factor it, holding model and image fixed, into a *naming* term (C0_named: a single slide, the candidate defect named, no reference, no forced-evidence gate) and a *pairing* term (the true 2-AFC against the clean twin), with a clean-vs-clean *guess-floor* control (Fig. 6). The answer is *defect-dependent* and consistent across all four capable models. For G7—the off-taxonomy render class C0 cannot *name*—**naming alone** recovers it (Qwen3.5-9B 0.50 $\rightarrow$ 0.93, Qwen3.6-27B 0.51 $\rightarrow$ 1.00, Qwen3.5-27B 0.79 $\rightarrow$ 1.00, Gemma4-31B 0.69 $\rightarrow$ 0.88; paired-clean McNemar Holm-corrected $p \leq 0.002$), and the paired reference adds nothing on top: this is *format suppression*, the model saw G7 all along. For G1/S6 the opposite holds: naming alone does *not* help (C0_named ≈ 0.50) and the recovery needs the paired clean reference (2-AFC $\rightarrow$ 0.97–1.00)—an *availability-of-reference* effect, properly scoped as such. Crucially, the guess-floor control rules out a forced-pick artifact for both: on two *clean* slides the capable models correctly answer “tie” 92–100% of the time and fabricate a consistent winner on $\leq 2\%$ of pairs, so the 2-AFC ceiling is real signal, not always-pick-one. The paper’s mechanism claim—*format suppression, not the eyes*—is thus established for the constructed G7 (and the C3 atomic recovery it mirrors); the declared G1/S6 forced-choice recovery is the milder availability-of-reference effect.

It transfers to real slides. On the human-annotated SLIDEAUDIT set, even with no structure available (so no linter), C3 beats C0 pointwise for every class—brand color 0.52 $\rightarrow$ 0.81, alignment 0.56 $\rightarrow$ 0.72—confirming the format effect is not a synthetic artifact. Absolute geometry on bare real pixels stays modest (0.55–0.74 outside blatant overlap): real fine geometry still needs the linter, which needs the element structure bare images lack. The hybrid’s full power requires native structure; on image-only real data it runs in a degraded VLM-only mode—a ceiling set by missing structure, not missing method.

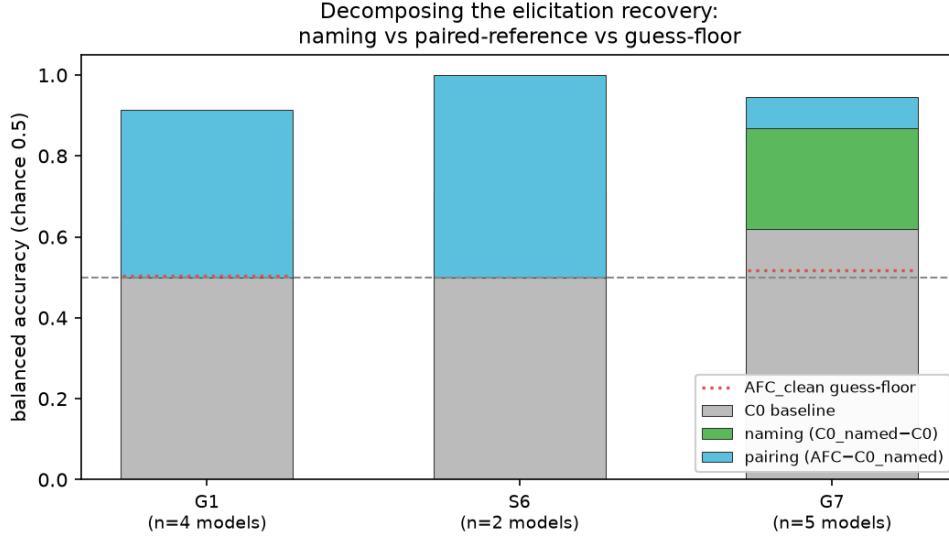


Figure 6: What drives the recovery: naming vs. a paired reference. Holding model and image fixed on the freeform set, the chance→ceiling jump factors into a *naming* component (C0_named-C0: one slide, the defect named, *no* reference) and a *pairing* component (AFC-C0_named: a clean twin supplied). For the constructed render class G7 naming alone carries it (green); for declared geometry G1 and semantic S6 the paired reference does (blue). The AFC_clean guess-floor (dotted) is near chance, so neither is a forced-pick artifact. Mean over the models that recover (AFC ≥ 0.7).

5.2 The minimal hybrid critic covers the most

Complementary coverage the diagnosis predicts. We evaluate five critics by *named attribution*: the correct defect must be emitted on the defective slide and not on its clean twin. The coverage is exactly what the diagnosis prescribes—declared geometry to the linter, the format-suppressed classes to a C3-elicited VLM. Re-eliciting the *same* VLM from C0 to C3 and adding the linter for declared geometry takes coverage from the conventional VLM-C0’s **2/9** (0.59) to **8/9** at **0.89**, matching the pre-registered routed hybrid (**8/9**, **0.88**) and beating either single engine (linter: **5/9** at 0.69; VLM-C3: **4/9** at 0.72; Fig. 7, Table 2). The nine classes are the image-level paired-clean classes; G4/S3 are linter-owned and S2/S5 are deck-scope (Sec. 6). The E3 control settles *why* this is the headline: per-class routed prompt/LLM tuning is *not* load-bearing here—*linter* $\oplus$ *C3-VLM* matches it—so the result is the complementary coverage the diagnosis predicts, not a tuned routing architecture.

A data-driven routing correction. We initially routed S1 title/body mismatch to the text LLM, but the text-only probe over-flags (0.25, precision 0.09) while the image-bearing VLM names it reliably (0.94, precision 0.90): title/body mismatch needs the *rendered layout*, not just the text. We re-route S1 to the VLM—and specifically to its *open* pointwise prompt (C0, 0.94) rather than the atomic C3 (0.83): recall is already saturated on S1 (both elicitations catch all 18 defectives), so the forced yes/no of C3 only adds false positives, dropping specificity 0.89→0.67. Atomic elicitation is therefore *not* a free lunch—it recovers recall-limited classes (G7, S4) but inflates false positives on subjective ones—which is itself the reason the critic routes the *elicitation* per class instead of forcing C3 everywhere. The one image-level class the deployable hybrid fails to cover is S6 image/text contradiction, which stays at 0.50 single-image for every engine—but the 2-AFC diagnosis (Fig. 4) shows the model *does* perceive it (it over-asserts pointwise rather than missing it; recall 1.0, specificity ~ 0), so the gap is availability-of-reference, not blindness—in pointed contrast to the genuine G6 page-offset blind spot.

5.3 G7: the linter-blind class that specialised rewards also miss

The construction. G7 is defined to make the linter’s blind spot falsifiable: a declared box that is legal (in-margin, non-overlapping) but whose rendered content overflows its container. The IR carries only legal short text; the overflow lives in the renderer, not the IR, so the linter and any structure-only model are blind by construction. A self-check confirms the geometry linter returns zero findings on 90/90 G7 defectives.

Table 2: Per-class named-attribution balanced accuracy with 95% Wilson intervals. E3 adds C3-everywhere and linter+VLM-C3: the minimal hybrid matches routed coverage and slightly exceeds its mean. n = paired positives with matched clean controls; precision for every cell is in the released reports.

Class	routed to	linter	VLM-C0	VLM-C3	linter+C3	routed	n
G1 overflow	linter	1.00 [.91,1]	0.50 [.46,.54]	0.50 [.46,.54]	1.00 [.91,1]	1.00 [.91,1]	40
G2 overlap	linter	0.80 [.68,.87]	0.71 [.60,.79]	0.86 [.74,.92]	0.80 [.68,.87]	0.80 [.68,.87]	40
G3 alignment	linter	0.93 [.85,.96]	0.62 [.55,.68]	0.71 [.63,.77]	0.93 [.85,.96]	0.93 [.85,.96]	70
G5 color	linter	1.00 [.91,1]	0.50 [.46,.54]	0.68 [.57,.75]	1.00 [.91,1]	1.00 [.91,1]	40
G6 margin	linter	1.00 [.90,1]	0.50 [.45,.55]	0.50 [.45,.55]	1.00 [.90,1]	1.00 [.90,1]	36
G7 render-overflow	VLM (C3)	0.00 [0,.54]	0.50 [.46,.54]	1.00 [.91,1]	1.00 [.91,1]	1.00 [.91,1]	40
S1 title/body	VLM (C0)	0.50 [.41,.59]	0.94 [.75,.98]	0.83 [.63,.92]	0.83 [.63,.92]	0.94 [.75,.98]	18
S4 density	LLM	0.50 [.45,.55]	0.51 [.44,.59]	0.93 [.81,.97]	0.93 [.81,.97]	0.72 [.60,.80]	36
S6 image/text	VLM	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	18
mean / classes covered		0.69 / 5/9	0.59 / 2/9	0.72 / 4/9	0.89 / 8/9	0.88 / 8/9	

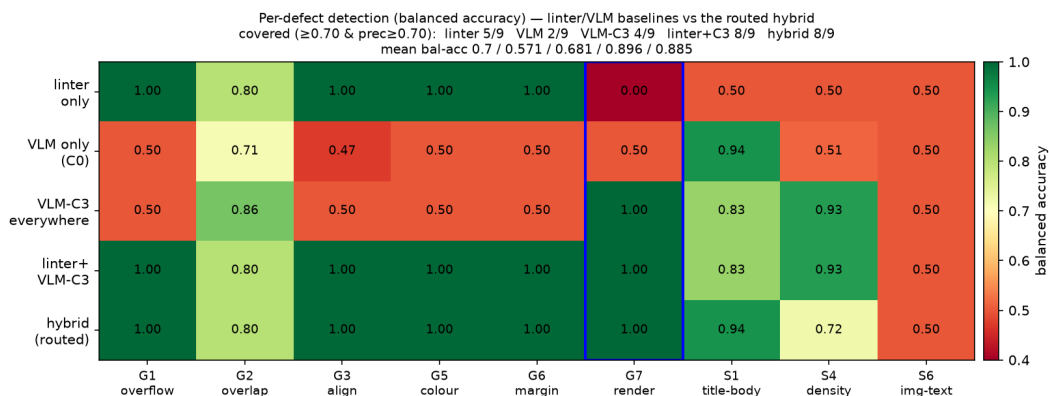


Figure 7: Coverage by engine. Per-defect balanced accuracy for the linter, VLM-C0, VLM-C3, linter+VLM-C3, and the routed hybrid. Minimal linter+C3 matches routed coverage; G7 (boxed) is the linter-blind cell rescued by C3.

What kind of reward sees G7? Perceptual capability, not the training-domain label. The natural rebuttal is “just train a neural reward model.” We test it directly by auditing **five** published reward scorers—now *two per audited category*, so the answer is category- not instance-level—on the same paired slides: does each score the clean slide above its defective twin? The scorers are a *document* reward (**DocReward** [13], Qwen2.5-VL-3B with a Bradley–Terry head over 117K paired documents); two *general*-multimodal rewards on *distinct* backbones (**Skywork-VL-Reward-7B** [14], a Qwen2.5-VL-7B value head, and **PickScore** [16], a CLIP-H/14 contrastive reward fine-tuned on 500K human choices); and two *aesthetic*/perceptual scorers (**LAION-Aesthetic** [15], a linear artistic-appearance head on CLIP-L/14, and **CLIP-IQA** [17], a zero-shot “Good/Bad photo” quality probe on the same CLIP). Each cleanly penalizes several in-taxonomy defects it can see (Table 3, Fig. 8), so each is a working scorer, not a domain mismatch. On G7 the picture splits—but *not* along the domain label. G7 is detected (95% CI above chance, $n=90$) by *both* general-multimodal rewards (Skywork-VL 0.79, [0.69, 0.86]; PickScore 0.71, [0.61, 0.80])—a *category-level* result across two unrelated backbones—and, tellingly, by the zero-shot perceptual-quality probe (CLIP-IQA 0.83, [0.74, 0.90]); it is missed (CI spans 0.5) by the symbolic linter (0.00), the document reward (DocReward 0.48, [0.38, 0.58]) and the artistic-aesthetic head (LAION 0.57, [0.46, 0.66]). So the dividing line is not “general vs. narrow domain” but *perceptual capability*: a scorer that reads generic rendered quality—a capable general-multimodal reward, or even a plain CLIP quality read—catches the visible overflow, whereas a head tuned to a *specialised* objective (document structure, artistic appeal) is blind to it even when, as for CLIP-IQA vs. LAION, the two share a backbone. This is the reward-side counterpart of the C0→C3 recovery: the perception is present; a specialised read-out suppresses it. The hybrid thesis is unchanged and now category-level: G7 needs a perceptually-capable engine, and that engine works whether it is *prompted* (the C3 recovery, 0.93–1.00) or carries a *reward head* (two general-mm rewards, 0.71–0.79)—a reviewer who tests either

Table 3: Multi-reward audit: paired preference accuracy $P(r_{\text{clean}} > r_{\text{def}})$ per class (chance = 0.5; G7 with 95% CI, $n=90$). **Five** published scorers, now *two per audited category* (general-multimodal: Skywork-VL + PickScore on *distinct* backbones; aesthetic: LAION + CLIP-IQA), so the G7 read is category- not instance-level. G7 (boxed in Fig. 8) is detected (CI above 0.5) by *both* general-mm rewards and by the zero-shot perceptual-quality probe (CLIP-IQA), and missed (CI spans 0.5) by the document reward and the artistic-aesthetic head—so detection tracks perceptual capability, not the training-domain label. $\checkmark/\times$ in the G7 column marks reliable detection. G5 is the internal-chromatic re-operationalisation of Sec. 4 (one element hue-swapped vs. its sibling column, $n=80$), a *visible* defect most scorers catch—not the linter-only blind spot G7 is.

Reward scorer	category	G1	G2	G5	G7	S4	S6
DocReward-3B	document	0.93	1.00	0.72	$\times$ 0.48 [.38, .58]	1.00	1.00
Skywork-VL-7B	general-mm	0.94	0.72	0.57	$\checkmark$ 0.79 [.69, .86]	0.92	1.00
PickScore-v1	general-mm	0.74	0.59	0.70	$\checkmark$ 0.71 [.61, .80]	1.00	0.72
LAION-Aesth.	aesthetic	0.37	0.91	0.78	$\times$ 0.57 [.46, .66]	0.75	0.61
CLIP-IQA	aesthetic	0.44	0.50	0.65	$\checkmark$ 0.83 [.74, .90]	0.50	0.72
n (paired)		54	54	80	90	36	18

will find detection. The full split, including the aesthetic scorers that disagree, is in Table 3. (Naming containment in Skywork’s prompt lifts G7 to 0.87, reinforcement rather than rescue. On G5—re-operationalised as an internal chromatic clash, one element hue-swapped against its aligned sibling column and so visible from the slide alone (Sec. 4)—most scorers now react (0.65–0.78; only Skywork is marginal at 0.57), exactly as the perceptual-capability reading predicts for a visible defect: a chromatic inconsistency is *not* the linter-only blind spot G7 is. The linter still owns G5 at 1.00, but its advantage there is exactness, zero false positives, and knowledge of the *declared* palette—not that the clash is imperceptible.)

Perturbation fidelity: 45% of injected geometry never renders. The reward audit ties to the template result of Sec. 4. Rendering every injected defective two ways—faithfully and snapped-to-master—we find 45% of injected geometry defects (100% of overlap, alignment, margin; 20% of overflow) become pixel-identical to their clean twin after snapping. Feeding the *snapped* pairs to *all five* reward scorers gives preference accuracy 0.00 at a reward gap of exactly 0.000: the two images are pixel-identical, so no critic—document, general-multimodal, or aesthetic—can react—a dataset-integrity hazard (identical inputs force identical scores), not a reward-model deficiency. This part *is* model-agnostic by construction, and any reward model trained or evaluated on snap-rendered images with IR-derived labels inherits these zero-signal pairs. Perturbation fidelity must be checked at the pixel level, not assumed from the IR—a prerequisite the field largely skips, and the reason our elicitation experiments render defectives faithfully.

6 The semantic engine: a specialist examiner

The hybrid’s semantic engine can be a zero-shot model, but a small specialist does better on its own ground. We finetune Qwen3-VL-8B with QLoRA on programmatically injected defects (2,142 routed SFT records: semantic pointwise, geometry restate-from-structure with image-only abstention, and G1/S6 pairwise), and evaluate on held-out splits.

An 8B examiner surpasses a zero-shot 30B on *in-distribution* semantics. On in-domain held-out page-semantic inspection, the finetuned 8B reaches balanced accuracy 1.000 on both S1 title/body and S4 density, beating zero-shot 8B (0.78 / 0.53) and zero-shot 30B (0.93 / 0.77) (Table 4). The largest gain is S4 density recall, 0.0 $\rightarrow$ 0.97, where the finetuned 8B (1.00) beats the zero-shot 30B (0.65) at two-proportion z , $p = 4.5 \times 10^{-7}$: an 8B examiner surpasses a 4 $\times$ larger zero-shot model on the examiner’s core job. The advantage is not a prompt-format artifact—re-running the zero-shot baselines at the finetuned model’s own trained prompt format collapses them to 0.50, so the format asymmetry favors the baselines, and the finetuned model still wins at each model’s best format. On held-out *severities* the semantic edge holds (0.89 vs. 0.50–0.59).

It abstains instead of hallucinating geometry. Under image-only inspection, the finetuned model stays at 0.50 on G2–G6 with *zero* false positives: trained with image-only abstention targets, it declines to hallucinate geometry from

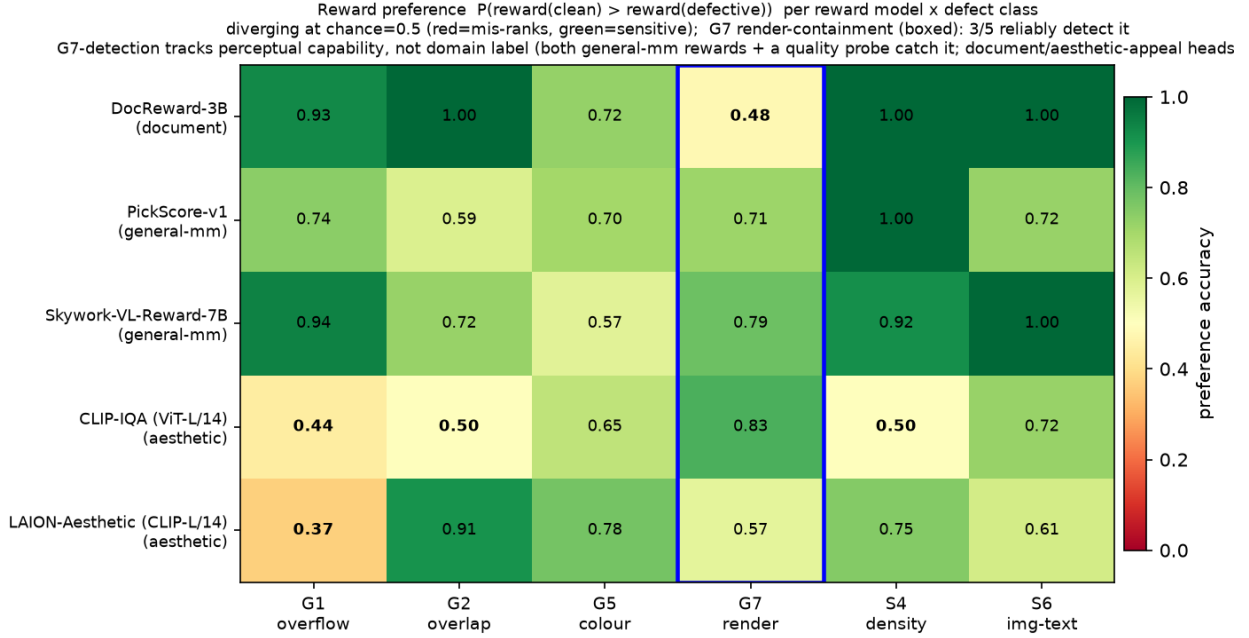


Figure 8: G7-detection tracks perceptual capability, not the training-domain label. Preference accuracy per reward model $\times$ defect class (green = sensitive, red = mis-ranks; chance = 0.5), five scorers, two per audited category. Every scorer is sensitive to several in-taxonomy classes, but for the boxed G7 render class detection splits by *capability*: both general-multimodal rewards (Skywork-VL 0.79, PickScore 0.71, distinct backbones) and the zero-shot perceptual-quality probe (CLIP-IQA 0.83) catch it, while the document reward (DocReward 0.48) and the artistic-aesthetic head (LAION 0.57) sit at chance—so the blind spot belongs to *specialised* critics, and is caught by any perceptually-capable scorer (prompted VLM, VL reward, or CLIP quality read).

pixels and leaves those classes to the linter—the designed behavior, and the reason the hybrid’s geometry column comes from the linter rather than the VLM.

Two limits on the reading. Two results bound an over-broad reading. First, *deck-scope* semantics fail pointwise: the finetuned model is degenerate on S2 (recall 1.0, specificity 0.0—always flags) because the SFT mix contained no clean-*deck* negatives, and it abstains on the held-out S5; deck-scope classes were never given the pairwise head the diagnosis prescribes for this failure mode, and fixing this needs a retrain we did not run. Second, there is a *sim2real* gap: on image-only SLIDEAUDIT, the finetuned model’s synthetic S4 strength (0.97) does not transfer (0.50); only the zero-shot 30B shows any real image-only density signal (recall 0.53 vs. the finetuned 0.075, $p = 1.1 \times 10^{-5}$). Scale, not finetuning, carries bare-pixel transfer—and the finetuned model learned a defect *distribution*, not a universal notion of slide quality. We make this concrete next.

7 External validity

A real pre-sales agent deck. We run the same five-level examiner gradient on a real 16-slide proposal produced by a deployed PowerPoint agent. Its verifiable defect is 20 unfilled template placeholders (“add a title here”) across 7 of 16 slides—a semantic-completeness defect the geometry linter is blind to by construction (the boxes are geometrically fine). The result reinforces the thesis from an unexpected angle: the zero-shot 30B is the best detector (recall 1.0, names the placeholder on 4/7 slides, scores them 0.03 vs. 0.64 for clean), while the *finetuned* 8B—top of the intrinsic-quality ranking—is among the *worst* on this out-of-distribution defect (names it 0/7), and the linter’s apparent recall is spurious geometry noise with zero placeholder mentions. The ranking on a novel defect is by general-VLM ability, not by the specialist’s in-distribution quality: the examiner learned a distribution, not universal quality. Feeding the flagged placeholders back to the agent’s own generator for one revision pass drives the verifiable defect from 20 to 0.

Table 4: Page-semantic inspection, balanced accuracy [95% Wilson] on paired clean (best modality). The finetuned 8B examiner beats both zero-shot baselines, including the 30B, *in-distribution*. Deck-scope S2/S5 are a negative result (text). n = paired positives (2-AFC rows = forced-choice pairs); the S6 2-AFC rests on only 12 pairs (wide interval).

class	ft-8B	zs-8B	zs-30B	n
S1 title/body	1.000 [.95,1]	0.778 [.69,.85]	0.933 [.87,.95]	36
S4 density	1.000 [.97,1]	0.529 [.50,.58]	0.774 [.69,.84]	60
G1 overflow (2-AFC)	1.000 [.97,1]	0.975 [.93,.99]	0.700 [.61,.77]	120
S6 image/text (2-AFC)	1.000 [.76,1]	1.000 [.76,1]	0.500 [.25,.75]	12
S2 narrative (deck)	0.500 [†] [.45,.53]	0.646 [.53,.72]	0.549 [.42,.66]	36
S5 missing-logic (deck)	0.500 [.47,.53]	0.500 [.47,.53]	0.567 [.45,.68]	60

[†] degenerate (always-flag): the SFT mix had no clean-deck negatives. See text.

The structured attribution holds on real layouts. The body runs the A/B/C attribution on synthetic slides; the cleanest discrimination needs it on *real* geometry with a *real* oracle, which seemed to require human or model annotation. We obtain it without either: ZENODO10K [18] ships real CC-licensed .pptx with intact element XML, so python-pptx extracts a geometry/text/style oracle *lossless with respect to the declared IR* straight from the source file (the IR→render gap that G7 exploits is exactly what this oracle does *not* capture, by design)—no human, no self-annotation bias. From 26 real decks we inject one single-shape defect per slide in *PPTX XML space* (overflow, overlap, alignment offset, font, margin) and render both the clean and the defective one-slide deck with the same real renderer (LibreOffice), so the paired pixel diff isolates the injected defect on real pixels (209 paired slides, 5 classes, 3 VLMs / 3 families). Two results follow. First, the §4 “snap absorbs 45%” hazard is *template-specific*: real free-form decks absorb only 7% (render-fidelity 0.93), so the perturbation is faithfully present—a positive control on that claim. Second, *all three* attribution outcomes appear on real geometry: coarse defects (overflow, overlap) are **image-sufficient** (balanced accuracy A 0.70–0.74, structure adds nothing); a margin bleed is a **perception bottleneck the oracle rescues** for the weaker VLMs (A 0.59 → B 0.70)—the exact “missing structure, not missing capability” cell; and fine alignment stays near chance in *every* channel for *every* model (A = B = C ≈ 0.50; the internal-contrast re-run at a supra-threshold 44 px offset reproduces this, A 0.54/B 0.51/C 0.52, $n=41$). This is *not* a hard capability floor—on clean synthetic slides the same alignment defect recovers under targeted/relative elicitation once it clears acuity (Fig. 3)—but on real cluttered layouts the misaligned element cannot be isolated among many genuine ones and specificity collapses, so for declared geometry the symbolic linter (which reads the coordinates) remains the reliable detector regardless (Fig. 9).

Open-world: structure recovered from pixels does not restore the linter. The hybrid’s symbolic engine needs the declared IR a .pptx carries; a third-party deck exported to PDF/PNG ships none, so the natural question is whether *recovering* element structure from the rendered pixels can re-arm the linter. We test it directly: a transformers-native document-layout detector (PP-DocLayoutV2) returns region boxes on each render, class-agnostic NMS removes the detector’s cross-class duplicate boxes (which would otherwise manufacture phantom overlaps), and the boxes—normalised into the IR coordinate frame—feed the *same* geometry linter at its shipped operating point. On the real-layout set (GT IR *and* render available, so the full ladder and a recovery-fidelity IoU coexist) the answer is a clean, pre-registered negative (Fig. 10): recovered structure leaves the linter at chance on G1/G3/G4 (overflow needs text capacity, alignment needs declared positions, font needs point sizes—none survive a pixel→box projection), and rescues only overlap partially (G2 balanced accuracy 0.54 vs. native-IR 0.67 vs. VLM-only 0.87). The detector *over-segments*—≈ 2× more boxes than the GT IR, recovery recall at IoU≥0.5 only 0.30–0.39—so genuine element collisions are split apart and overlap recall falls 0.78 → 0.29. The same picture holds on image-only SLIDEAUDIT (no IR at all): the recovered linter clears chance only on G2 (0.55) while the C3-elicited VLM reaches 0.94. Two points sharpen the deployment scope. First, on real third-party decks even *native* IR is weakened—the decks carry no expected-position/font bookkeeping, so the linter’s G3/G4 rules are also at chance, and its margin/overlap rules *over-fire* on tight real layouts (specificity 0.11/0.56), eroding the near-zero-false-positive advantage it enjoys on clean synthetic IR. Second, as a direct consequence the VLM-only engine *dominates every coarse class* on real decks. The linter’s edge is an in-distribution, clean-native-IR phenomenon; open-world—recovered or not—the deployable critic is VLM-bound. This bounds the contribution without touching the diagnostic thesis: the full symbolic–neural critic is for IR-owning generation agents, and for bare third-party pixels the critic is the C3-elicited VLM.

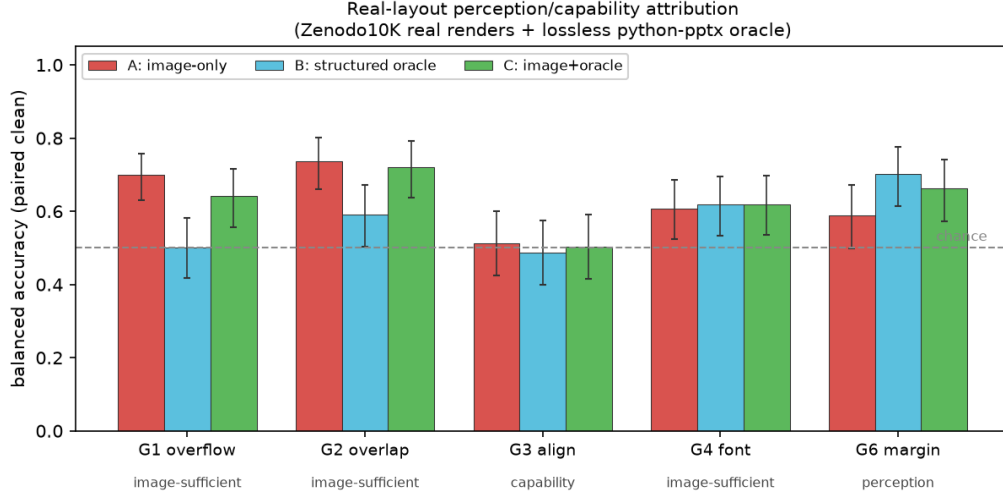


Figure 9: The perception/capability split reproduces on real layouts (ZENODO10K real renders + lossless python-pptx oracle; mean over 3 VLMs / 3 families, balanced accuracy on paired clean; error bars = 95% Wilson on counts pooled across models, $n=76\text{--}90$ paired per class). Coarse defects (G1 overflow, G2 overlap) are image-sufficient; G6 margin is a *perception* bottleneck the structured oracle rescues ($B>A$); G3 alignment stays near chance in every channel on these real cluttered decks ($A=B=C\approx 0.5$)—not a hard capability floor (it recovers on clean slides above threshold, Fig. 3) but unreliable enough on real layouts that it must route to the linter. Intervals on G3/G4/G6 include chance, consistent with their small per-class n (see released reports).

8 Limitations

We *do* run the structured (image+oracle) evaluation on real slides that separates “missing structure” from “missing capability” on real geometry (§7): a lossless `python-pptx` oracle over real CC-licensed decks sidesteps both the annotation labour and the self-annotation bias, and the split reproduces—fine alignment stays near chance in every channel on real cluttered decks (it recovers on clean slides above a magnitude threshold, but real clutter defeats it) while a margin bleed is structure-rescuable. Its scope is bounded: the defects are *injected* (real geometry, real renderer, real oracle, but not naturally-occurring), so the third-party SLIDEAUDIT set (image-only, naturally-defective) remains our complementary real probe. We audit five published reward scorers, two per audited category (general-multimodal, aesthetic), with the document category at one (a second document/design reward, PosterReward, needs an ms-swift `seq_cls` runtime we did not stand up, as does an InternLM-family reward whose PLoRA image path failed to load); the category-level reads are therefore solid for general-multimodal and aesthetic and instance-level for document. Our attribution protocol shares its core with concurrent perception/reasoning work, differentiated by domain, the lossless oracle, and the loop to a critic. The downstream effect is bounded on both sides: examiner quality correlates +0.66 with self-refine gain under a strong generator but at sub-1% magnitude (it floors most briefs at the first draft), and re-running the identical gradient with an unfloored weak generator ($1.43\times$ the headroom) collapses that correlation to 0.00. A controlled actionability A/B (generator, task, and seed fixed) locates this null as an axis mismatch, not a revision incapacity—the weak generator’s headroom is *coverage* (deck completeness), which a perception/layout critic does not address (the geometry critique moves quality by $\Delta q=+0.00$ against an explicit coverage critique’s $\Delta q=+0.32$)—so the examiner’s verifiable evidence is its intrinsic and case-study performance, not this loop.

Two further caveats temper the central claim. First, the forced-choice and atomic-binary elicitations supply a contrastive reference and a named target that the pointwise format withholds, so one must ask whether the $0.50 \rightarrow 1.00$ recovery is *elicitation* or *reduced task difficulty*. A clean ablation settles it (the single-slide, defect-type-named *absolute* condition C0_named, plus a clean-vs-clean guess-floor control; Sec. 5.1, Fig. 6): the answer is defect-dependent—for the constructed render class G7 naming alone recovers detection (format suppression), while for declared G1/S6 the recovery genuinely requires the paired reference (availability-of-reference). We therefore make the strong mechanism claim only for G7 and its C3 mirror, and label the G1/S6 forced-choice effect as reference-availability; the guess-floor control (capable models answer “tie” on 92–100% of clean pairs) rules out a forced-pick artifact for both. Second, the 8/9 coverage (0.89 for linter+VLM-C3; 0.88 for the pre-registered routed hybrid) is a *native-structure* number; on image-only real decks (no IR for the linter) the hybrid degrades to its VLM-only mode, so that figure is an upper bound

E5 — open-world hybrid: structure recovered from pixels does not restore the symbolic linter (the VLM dominates coarse geometry)

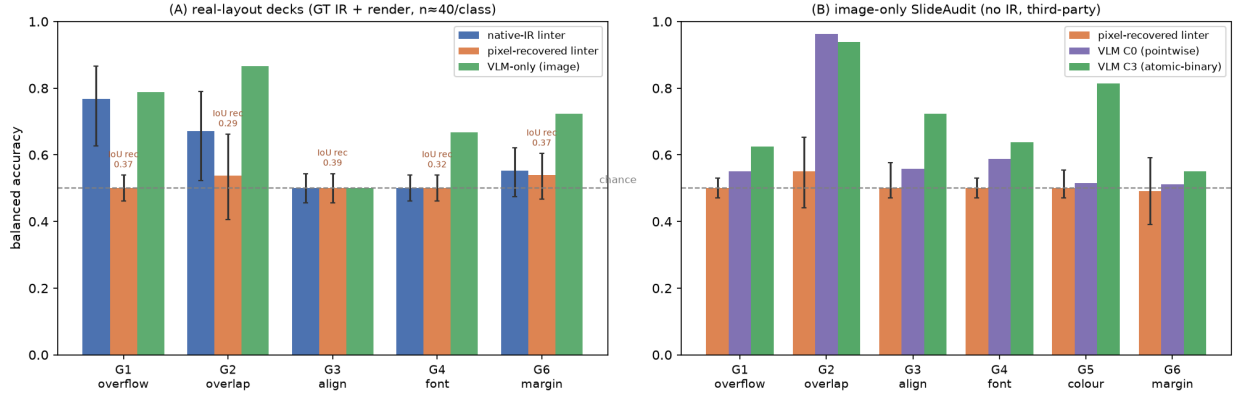


Figure 10: Open-world hybrid (E5): structure recovered from pixels does not restore the symbolic linter. A document-layout detector (PP-DocLayoutV2) recovers element boxes from the render; the *same* geometry linter runs on them. **(A)** real-layout decks (GT IR + render, $n \approx 40$ /class): native-IR linter vs. pixel-recovered linter vs. VLM-only floor, Wilson CIs; recovered structure rescues only overlap (G2) and is silent on fine geometry, while the VLM dominates every class. **(B)** image-only SLIDEAUDIT (no IR, third-party): the recovered linter clears chance only on G2; the C3-elicited VLM is far stronger. The linter’s near-zero-FP advantage requires clean native IR.

for IR-owning generation agents, not an open-world guarantee. Recovering element structure from the pixels with a layout detector and re-running the linter does *not* close the gap (§7, Fig. 10)—it rescues only overlap, partially, and the VLM-only engine dominates every coarse class on real decks, so the open-world critic is VLM-bound.

9 Conclusion

A slide-defect critic should not be one model. Telling apart the two blindnesses—a magnitude-gated acuity floor versus format-suppressed perception—turns a vague “VLMs are bad at layout” into an actionable routing rule: rules to the linter and perceivable-but-suppressed classes to a VLM under an elicitation that the diagnosis proves recovers them. The E3 control makes the rule simpler: linter plus one C3-prompted VLM covers 8/9 classes, matching the pre-registered routed hybrid and beating either single engine. A constructed render class G7, audited over five published rewards (two per category), shows that the linter’s blind spot is shared by a document reward and an artistic-aesthetic head but *not* by either of two general-multimodal rewards or a perceptual-quality probe—so G7-detection tracks perceptual capability, not the training-domain label, and G7 needs a perceptually-capable engine, prompted or reward-headed, not a specialised one. The recovery replicates across four of the six VLMs tested and four families and transfers to real slides, which makes the central claim a property of the *elicitation* $\times$ *defect* interaction rather than of any one model. The unit of contribution is the lossless-oracle diagnosis—which bottleneck a defect hits, and therefore which engine it needs—and the falsifiable class that stress-tests it; the routed critic is the consequence, not the headline.

Reproducibility

All analysis scripts, the defect injector, the two linters, the elicitation harness, the hybrid router, the multi-reward audit adapters, the real-layout (ZENODO10K) IR/inject/render pipeline, and the per-Part reports are released. Experiments ran on 4xRTX 3080 (20 GB) with vLLM-served open VLMs (Qwen3.5/3.6, Gemma4, InternVL3.5, Ovis2.5), a QLoRA-finetuned Qwen3-VL-8B, and five published reward scorers (DocReward-3B; Skywork-VL-Reward-7B and PickScore; a LAION aesthetic predictor and CLIP-IQA) behind a uniform adapter; serving recipes (tensor-parallel, AWQ, disabled thinking, fp8-KV caveats on Ampere) are in the repository. Weights, renders, and raw rollouts are not redistributed. Every balanced-accuracy cell in the tables and figures carries its 95% Wilson interval and per-cell n ; the full family of paired tests is reported with Holm (FWER) and Benjamini–Hochberg (FDR) multiplicity correction over all 61 reported significance tests (released multiplicity report), and headline claims are stated as surviving or downgraded under that correction.

References

- [1] On the Perception Bottleneck of VLMs for Chart Understanding. arXiv:2503.18435.
- [2] Hidden in Plain Sight: VLMs Overlook Their Visual Representations. arXiv:2506.08008.
- [3] Tong et al. Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs. CVPR 2024; arXiv:2401.06209.
- [4] Your Reasoning Benchmark May Not Test Reasoning: Revealing Perception Bottleneck in Abstract Reasoning Benchmarks. arXiv:2512.21329.
- [5] LED Benchmark: Diagnosing Structural Layout Errors for Document Layout Analysis. arXiv:2507.23295.
- [6] VLM Judges Can Rank but Cannot Score: Task-Dependent Uncertainty in Multimodal Evaluation. arXiv:2604.25235.
- [7] Liusie, Manakul, Gales. LLM Comparative Assessment: Zero-shot NLG Evaluation through Pairwise Comparisons using Large Language Models. EACL 2024; arXiv:2307.07889.
- [8] Tripathi, Wadhwa, Durrett, Niekum. Pairwise or Pointwise? Evaluating Feedback Protocols for Bias in LLM-Based Evaluation. arXiv:2504.14716.
- [9] AeSlides: Incentivizing Aesthetic Layout in LLM-Based Slide Generation via Verifiable Rewards. arXiv:2604.22840.
- [10] Presenting a Paper is an Art: Self-Improvement Aesthetic Agents for Academic Presentations (EvoPresent/PresAesth). arXiv:2510.05571.
- [11] VLM-SlideEval: Evaluating VLMs on Structured Comprehension and Perturbation Sensitivity in PPT. arXiv:2510.22045.
- [12] SlideAudit: A Dataset and Taxonomy for Automated Evaluation of Presentation Slides. arXiv:2508.03630.
- [13] DocReward: A Document Reward Model for Structuring and Stylizing. arXiv:2510.11391.
- [14] Skywork-VL Reward: An Effective Reward Model for Multimodal Understanding and Reasoning. arXiv:2505.07263; weights Skywork/Skywork-VL-Reward-7B.
- [15] LAION-AI. Aesthetic Predictor: a linear estimator on CLIP for image aesthetic quality (LAION-Aesthetics). Software, 2022. <https://github.com/LAION-AI/aesthetic-predictor>.
- [16] Kirstain et al. Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation. NeurIPS 2023; arXiv:2305.01569; weights yuvalkirstain/PickScore_v1.
- [17] Wang, Chan, Loy. Exploring CLIP for Assessing the Look and Feel of Images. AAAI 2023; arXiv:2207.12396.
- [18] PPTAgent: Generating and Evaluating Presentations Beyond Text-to-Slides (introduces the Zenodo10K corpus of real .pptx). arXiv:2501.03936; dataset Forceless/Zenodo10K.